

Appendix 24

Projection in the lowest Landau level

Every one-electron wave function $\psi(z)$ in the LLL has the form

$$\psi(z) = f(z)e^{-\frac{|z|^2}{4\ell^2}}, \quad (\text{A24.1})$$

where $f(z)$ is an analytic function of z . Conversely, every analytic function $f(z)$ determines a one-electron wave function in the LLL. Operators that depend on the coordinates x and y can be expressed in terms of z and z^* . It is evident that $z\psi(z)$ belongs to the LLL if $\psi(z)$ does, but what about $z^*\psi(z)$? To answer this question consider the “matrix element” of z^* between two states in the LLL:

$$\int g(z^*)z^*f(z)e^{-\frac{|z|^2}{2\ell^2}}dxdy. \quad (\text{A24.2})$$

Writing $z^*e^{-\frac{|z|^2}{2\ell^2}}$ as $2\ell^2\frac{\partial}{\partial z}e^{-\frac{|z|^2}{2\ell^2}}$ and integrating by parts we see that this integral can be rewritten as

$$\int g(z^*)\left[2\ell^2\frac{\partial}{\partial z}f(z)\right]e^{-\frac{|z|^2}{2\ell^2}}dxdy. \quad (\text{A24.3})$$

Thus the action of z^* within the LLL is equivalent to that of $2\ell^2\frac{\partial}{\partial z}$ acting only on the analytic part of the wave function ($f(z)$ in this case). This suggests that operators that are functions of z and z^* are projected in the LLL simply by the replacement $z^* \rightarrow 2\ell^2\frac{\partial}{\partial z}$. However, because z and the projected z^* do not commute with each other, one must still pay attention to their relative order. By considering, for example, the matrix element of zz^* between two LLL wave functions, we see that

$$\int g(z^*)zz^*f(z)e^{-\frac{|z|^2}{2\ell^2}}dxdy = \int g(z^*)\left[2\ell^2\frac{\partial}{\partial z}zf(z)\right]e^{-\frac{|z|^2}{2\ell^2}}dxdy. \quad (\text{A24.4})$$

Thus, the operator $\frac{\partial}{\partial z}$ must be placed on the left of all the z 's. This leads us to the final form of the prescription (Girvin and Jach, 1984): to project in the LLL an operator that is a function of z and z^*

- (i) Expand the operator in a power series of z and z^* ;
- (ii) Put all the z^* 's on the left of all the z 's;
- (iii) Replace z^* by $2\ell^2\frac{\partial}{\partial z}$;

- (iv) Remember that $\frac{\partial}{\partial z}$ operates only on the polynomial part of the wave function (*not* on the exponential part).

As a simple application, let us work out the relationship (10.170) between the projected structure factor and the full structure factor. Using the prescriptions given above, together with the disentanglement lemma (9.80) we see that

$$\begin{aligned}\hat{n}_{\vec{q}}\hat{n}_{-\vec{q}} &= \sum_j e^{-iq\ell^2 \frac{\partial}{\partial z_j}} e^{-i\frac{q^* z_j}{2}} \sum_k e^{+iq\ell^2 \frac{\partial}{\partial z_k}} e^{+i\frac{q^* z_k}{2}} \\ &= \sum_{j,k} e^{-iq\ell^2 \left[\frac{\partial}{\partial z_j} - \frac{\partial}{\partial z_k} \right]} e^{-iq^* \left[\frac{z_j - z_k}{2} \right]} e^{-\frac{q^2 \ell^2}{2} \left[\frac{\partial}{\partial z_k}, z_j \right]} \\ &= \overline{\hat{n}_{\vec{q}}\hat{n}_{-\vec{q}}} + N \left[e^{-\frac{q^2 \ell^2}{2}} - 1 \right].\end{aligned}\tag{A24.5}$$

Eq. (10.170) follows immediately from this and the fact that

$\langle \psi_0 | \hat{n}_{\vec{q}}^\dagger \hat{n}_{\vec{q}} | \psi_0 \rangle = \langle \psi_0 | \overline{\hat{n}_{\vec{q}}^\dagger \hat{n}_{\vec{q}}} | \psi_0 \rangle$. This is a fine example of the general property that the LLL projection of the product of two operators differs from the product of the individual projections.